

CSE213 CRA Report

Summer 2026

Topic: Simulating operations of a Private power generation company such as Summit Power		
ID: 2310604	Name: Ridwan Hasan Khandakar	User1: Accountant
		User2: BPDB Officer
ID: 2022752	Name: Kazi Fatema Tuj Johra	User3: HR Manager
		User4: Inventory Manager
ID: 2110304	Name: Rubama Mehnaj	User5: CEO
		User6: Plant Manager
ID: 2431072	Name: Samia Islam	User7: Engineer
		User8: Grid Operator

NOTE:

- Download this template and create a google-docs file with the following **file name format**:
 - MS1_GroupNo_MS1Version1_topicKeyword_id1_id2_id3_id4
 - Ex: MS1_Group90_MS1Version1_IRAS_1234567_2345678_3456789_4567890
- Group Leader will create the google-docs using this template and will share the google-docs file among group members. All of the group members need to work on the same document.
- Group leader must also upload the "viewer share" link of your google-docs file, so that the faculty can view your document anytime and check your progress (link will be given, soon).
- **ONLY ONE SUBMISSION from EACH GROUP**
- **To lock your content, you MUST upload a PDF file. NO LINK will be accepted.**
 - **By 04-June-26**, Saturday, the group leader MUST upload the PDF of your google-docs containing workflows for ALL 64 goals.
 - Among 64 workflows (of the goals) for 8 users, 16 workflows (4 workflows per member, 2 workflows/user), MUST look complete and comprehensive.
 - Submission Link will be given before the deadline.
- **If you fail to submit version-1 of your MileStone-1 by 04-June-26, you will not get any mark for MileStone-1**

Possible event type:

- UIE -- user input to trigger event
- UID -- user input to be considered as data
- OP – display content (output)
- VL – validation check (usually for user input)
- VR – verification check (verify with existing application data)
- DP – fetching data from file system and process it to get some calculated outcome (data processing)

NOTE:

there can be workflows for common **processes** as follows (**BUT, these are NOT GOALS**):

Process-1: Login			
			event type
Workflow:	event-1	Enter username(must be 4 digit number) and password	VL
	event-2	Validate login credentials against database	VL
	event-3	Verify user role and permissions	VR
	event-4	If login is successful, Open dashboard(role)	OP

Process-2: SignOut			
			event type
Workflow:	event-1	Button click for signout	UIE
	event-2	Take user back to login page	UIE

Process-3: SignUp (for example)			
			event type
Workflow:	event-1	Button click for signup	UIE
	event-2	Enter first and last name	UID
	event-3	Create username	UID
	event-4	Create password	UID
	event-5	Re-enter password	VR
	event-6	Validates that passwords match	VR

	event-7	Show success message: “SignUp successful”	OP
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Process-4: Change Password

			event type
Workflow:	event-1	Click “Change Password” button from dashboard	UIE
	event-2	Enter old password	UID
	event-3	Enter new password	UID
	event-4	Re-enter new password for confirmation	UID
	event-5	Validates that new passwords match	VL
	event-6	Verify old password with database	VR
	event-7	If valid, update password in user record	DP
	event-8	Show success message: “Password updated”	OP

Workflow of user specific goals:

User-1 name: Accountant

User-1, Goal-1	Description of Goal-1: Log income from electricity sales		event type
Workflow:	event-1	Login process	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks on “Sales Income” module	UIE
	event-4	System loads sales entry form with fields: Client/Distribution Company Name	OP

		- Electricity units sold (MWh) - Revenue amount - Date -	
	event-5	Accountant fills in sales information	UID
	event-6	Accountant selects transaction date	UID
	event-7	Accountant clicks “Log Sale” button	UIE
	event-8	System validates all fields and checks numeric format	VL
	event-9	If valid, system stores income record	DP
	event-10	Confirmation message: “Electricity sale logged successfully”	OP
	event-11	Entry appears in sales history table	Dp
User-1, Goal-2	Description of Goal-2: Track fuel and operational expenses		
Workflow:	event-1	login process	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Expenses” module	UIE
	event-4	System loads expense form with fields: - Expense category (Fuel / Maintenance / Utility / Other) - Amount - Date	OP
	event-5	Accountant selects expense type	UID
	event-6	Accountant enters amount and selects date	UID
	event-7	Accountants clicks “Add Expense”	UIE
	event-8	System validates input values	VL

	event-9	System saves expense record	DP
	event-10	Confirmation message : “Expense logged successfully”	OP
	event-11	Expense entry appears in monthly summary table	DP
User-1, Goal-3	Description of Goal-3: Manage salary payments		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Salary Management”	UIE
	event-4	System displays employees salary table: - Employee Name - ID - Salary - Payment Status	DP
	event-5	Accountant selects month from dropdown	UID
	event-6	System filters salary records for selected month	DP
	event-7	Accountant selects unpaid salary entries	UIE
	event-8	Accountant clicks “Process Payment”	UIE
	event-9	System verifies selected records	VR
	event-10	System updates salary payments status	DP
	event-11	Confirmation message : “Salaries processed successfully”	OP
User-1, Goal-4	Description of Goal-4: Generate monthly profit/loss reports		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Reports” module	UIE
	event-4	System displays month/year selection dropdown	OP

	event-5	Accountant selects reporting period	UID
	event-6	System validates selected period	VL
	event-7	System retrieves: - Sales income - Expenses - Salary costs - Fuel costs	DP
	event-8	System calculates profit/loss	DP
	event-9	Profit/Loss summary displayed with breakdown	OP
User-1, Goal-5	Description of Goal-5: Monitor fuel inventory valuation		
Workflow:	event-1	Login Process	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Fuel Inventory Valuation” module	UIE
	event-4	System displays inventory table: - Fuel type - Quantity - Unit cost	DP
	event-5	Accountant selects category/date range	UID
	event-6	System filters inventory records	DP
	event-7	System calculates total value: Quantity x Unit cost	DP
	event-8	Total valuation displayed	OP
User-1, Goal-6	Description of Goal-6: Approve high-cost purchase requests		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Purchase Approvals” module	UIE

	event-4	System loads pending purchase request list: • Cost • Description • Department	DP
	event-5	Accountant reviews requests	OP
	event-6	Accountant selects request	UID
	event-7	Accountant clicks “Approve” or “Reject”	UIE
	event-8	System verifies amount threshold	VR
	event-9	System updates request status	DP
	event-10	Confirmation message shown	OP
	event-11	Purchase history table updated	DP
User-1, Goal-7	Description of Goal-7: Handle tax and audit records		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Tax & Audit Records” module	UIE
	event-4	System displays categorized records: • Tax Returns • VAT • Audit Logs	DP
	event-5	Accountant selects record type/year	UID
	event-6	System filters records	DP
	event-7	Accountant selects a record	UIE
	event-8	System displays record details	OP
User-1, Goal-8	Description of Goal-8: Generate financial statements		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP

	event-3	Accountant clicks “Financial Statements” module	UIE
	event-4	System displays statement type dropdown: • Balance Sheet • Income Statement • Cash Flow Statement	OP
	event-5	Accountant selects statement type and reporting period	UID
	event-6	System validation selection	VL
	event-7	System retrieves: • Income records • Expense records • Tax data • Asset data	DP
	event-8	System generates formatted financial statement	DP
	event-9	Statement preview displayed	OP

User-2 name: BPDB Officer			
User-2, Goal-1	Description of Goal-1: Submit Power Demand Request		event type
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load BPDB Officer dashboard	OP
	event-3	Click “Power Demand Request” module	UIE
	event-4	System loads demand request form	OP
	event-5	Enter required power capacity (MW)	UID
	event-6	Select supply period from dropdown	UID
	event-7	Enter additional remarks	UID
	event-8	Click “Submit Request” button	UIE
	event-9	System validates entered information	VL
	event-10	System stores request in database/file	DP
	event-11	Confirmation message displayed	OP
User-2, Goal-2	Description of Goal-2: View Power Supply Agreements		
Workflow:	event-1	Login process execution	See login process
	event-2	Load BPDB dashboard	OP
	event-3	Click "Supply Agreements" module	UIE
	event-4	System retrieves agreement records	DP

	event-5	Display agreements in TableView	OP
	event-6	Select displays agreement record	UIE
	event-7	System displays agreement details	OP
	event-8	Review agreement information	DP
	event-9	Click “Download Agreement” button	UIE
	event-10	System generates downloadable file	DP
User-2, Goal-3	Description of Goal-3: Monitor Daily Power Supply		
Workflow:	event-1	Login process execution	See login process
	event-2	Load dashboard	OP
	event-3	Open "Power Supply Monitoring" module	UIE
	event-4	System retrieves daily supply records	DP
	event-5	Displays supplied power data in table	OP
	event-6	Select a date range	UID
	event-7	System filtered records	DP
	event-8	Calculated total supplied power	DP
	event-9	Displays summary information	OP
	event-10	Save monitoring act	UID
User-2, Goal-4	Description of Goal-4: View Invoices and Payment Records		
Workflow:	event-1	Login process execution	See login process
	event-2	Load dashboard	OP
	event-3	Click “Invoice & Payments” module	UIE

	event-4	Retrieve invoice records	DP
	event-5	Displays invoice records	DP
	event-6	Selects invoice record	UIE
	event-7	View payment details	OP
	event-8	Check payment details	DP
	event-9	Click “Download Invoice” button	UIE
	event-10	System generates invoice file	DP
User-2, Goal-5	Description of Goal-5: Submit Power Supply Complaint		
Workflow:	event-1	Events of login process to be executed	See login process
	event-2	Load dashboard	OP
	event-3	Click “Submit Complaint” module	UIE
	event-4	System loads complaint form	OP
	event-5	Select complaint category	UID
	event-6	Enter complaint details	UID
	event-7	Select affected supply date	UID
	event-8	Click “Submit” button	UIE
	event-9	System validates information	VL
	event-10	Complaint stored in records	DP
	event-11	Confirmation displayed	OP
User-2, Goal-6	Description of Goal-6: Request Additional Power Supply		
Workflow:	event-1	Events of login process to be executed	See login

			process
	event-2	Load dashboard	OP
	event-3	Open “Additional supply Request” module	UIE
	event-4	Enter required additional capacity	UID
	event-5	Select requested date range	UID
	event-6	Enter request justification	UID
	event-7	Clicks “Submit Request”	UIE
	event-8	Validate request information	VL
	event-9	Store request record	DP
	event-10	Notify Plant Manager	OP
User-2, Goal-7	Description of Goal-7: View Generation and Supply Reports		
Workflow:	event-1	Events of login process to be executed	See login process
	event-2	Load dashboard	OP
	event-3	Click “Reports” module	UIE
	event-4	Select report type	UID
	event-5	Select reporting period	UID
	event-6	System retrieves report data	DP
	event-7	Generate report summary	DP
	event-8	Displays report in table/chart	OP
	event-9	Click “Export Report”	UIE
	event-10	Download report file	DP
User-2, Goal-8	Description of Goal-8: Review Contract Performance		
Workflow:	event-1	Events of login process to be executed	See login

			process
	event-2	Load dashboard	OP
	event-3	Open “Contract Performance” module	UIE
	event-4	Retrieves contract records	DP
	event-5	Displays contract list	OP
	event-6	Select contract	UIE
	event-7	System calculates fulfillment statistics	DP
	event-8	Displays performance indicators	OP
	event-9	Generate performance summary	DP
	event-10	Export summary report	UIE

User-3 name: **HR Manager**

User-3, Goal-1	Description of Goal-1: Add New Employee		event type
Workflow:	event-1	Events of login process to be executed	See login process
	event-2	If login is successful, load "HR Manager" dashboard	OP
	event-3	Select "Add New Employee" option	UIE
	event-4	Fillup employee details: - First Name - Last Name - Date of Birth - Phone Number - Address - Joining Date - NID Number - Gender - Position - Marital Status - Religion	UID
	event-5	Verify if the provided information or data is valid	VL
	event-6	if verification of event-5 is successful: - Generate new employee ID - Save employee information - Show "Employee Created!" notification else: - Show "Invalid Data!" warning - goto event-4	DP, OP

User-3, Goal-2	Description of Goal-2: Remove any current employee		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Remove Current Employee” option	UIE
	event-4	Enter employee ID	OP
	event-5	Verify if provided data is valid	VR, VL
	event-6	If verification of event-5 is successful: - Show employee information Else: - Show “No match!” - Goto event-4	OP
	event-7	Show employee information	OP
	event-8	Show “Are you sure you want to remove this employee?”	OP
	event-9	User chooses “Yes” or “No” option	UIE
	event-10	If user selects “Yes”: - Remove employee information - Show “Employee Removed!” Else: Goto event-4	OP
User-3, Goal-3	Description of Goal-3: Update employee information		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Update Employee Information” option	UIE
	event-4	Enter employee ID	OP
	event-5	Verify if provided data is valid	VR, VL
	event-6	If verification of event-5 is successful: - Show employee information	OP

		Else: - Show "No match!" - Goto event-4	
	event-7	Update employee details: - Position - Phone Number - Salary - Address - Marital Status - Gender - Religion	UID
	event-8	Verify if the provided data is valid	VL
	event-9	If verification of event-6 is successful: - Save updated information - Show "Employee Information Updated" Else - Show "Invalid Data" warning - Goto event-5	OP, DP
User-3, Goal-4	Description of Goal-4: Employee attendance and performance grading		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "HR Manager" dashboard	OP
	event-3	Select "Employee Attendance and Performance Grading" option	UIE
	event-4	Enter employee ID	UID
	event-5	Verify if provided data is valid	VR, VL
	event-6	If verification of event-5 is successful: - Show employee information Else: - Show "No match!" - Goto event-4	OP
	event-7	Fillup attendance attendance and grading details: - Total days present - Total overtime hours - Operational efficiency score out of 10	UID
	event-8	Verify if the provided data is valid	VL

	event-9	If verification of event-7 is successful: - Calculate overall performance grade - Save attendance and grading data - Show "Performance Graded!" Else - Show "Invalid Data" warning - Goto event-5	DP, OP
User-3, Goal-5	Description of Goal-5: Search Employee Information		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "HR Manager" dashboard	OP
	event-3	Select "Search Employee Information" option	UIE
	event-4	Enter employee ID	UID
	event-5	Verify if employee exists	VR
	event-6	If verification of event-5 is successful: - Show employee information Else - "Employee Doesn't Exist" - Goto event-4	OP
User-3, Goal-6	Description of Goal-6: Create Notice		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "HR Manager" dashboard	OP
	event-3	Select "Create Notice" option	UIE
	event-4	Enter notice details: - Subject - Content - Expiry date	UID
	event-5	Verify if provided information is valid	VL
	event-6	If verification of event-5 is successful: - Show notice on dashboard - Show "Notice created!" notification	OP

		Else: - Goto event-4	
User-3, Goal-7	Description of Goal-7: Generate Salary		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Generate Salary” option	UIE
	event-4	Select target Month, Year and Department	UID, UIE
	event-5	For each current employee calculate salary based on: - Basic salary - Bonus - Overtime - Tax deduction	DP
	event-6	Generate salary slip	DP
	event-7	Show “Salary Generated!” notification	OP
User-3, Goal-8	Description of Goal-8: Candidate Selection		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Candidate Selection” option	UIE
	event-4	Select Job Circular ID	UIE, UID
	event-5	Shows applicant list and their test scores	OP
	event-6	For each applicant: - Fill up interview scores - Change status to “Selected” or “Rejected” - If user selects “Selected” - Sends automated job offer email - Else: - Send not hired email	UIE, UID, DP

User-4 name: **Inventory Manager**

User-4, Goal-1	Description of Goal-1: Add Equipment		event type
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Restock Equipment" option	UIE
	event-4	Fillup equipment details: - Name - Category - Quantity - Unit Price - Vendor	UID
	event-5	Verify if the provided product data is valid	VR, VL
	event-6	if verification of event-5 is successful: - Generate new product ID - Save product information - Show "Product Added!" notification else: - Show "Invalid Data!" warning - goto event-4	DP, OP
User-4, Goal-2	Description of Goal-2: Remove Damaged Equipment		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Remove Damaged Equipment" option	UIE
	event-4	Enter equipment ID	UID, UIE
	event-5	Verify if provided data is valid	VI, VR
	event-6	If verification of event-5 is successful: - Show "Are you sure you want to remove this product?"	OP

	event-7	Uses chooses “Yes” or “No” option	UIE
	event-8	If user selects “Yes”: - Show “Product Removed!” notification Else: - Goto event-4	DP,OP
User-4, Goal-3	Description of Goal-3: Parts Issuance		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “Inventory Manager” dashboard	OP
	event-3	Select “Parts Issuance” option	UIE
	event-4	Fill up issuance details: - Work order ID - Requester employee ID - Equipment ID - Quantity required	UID
	event-5	Verify if the requested equipment ID exists and if the entered quantity is currently available in stock	VR, VL
	event-6	If verification of event-5 is successful: - Goto event-7 Else: - Show “Insufficient stock! Available quantity [X]” warning - Goto event-4	OP
	event-7	Deduct the issued quantity from the inventory database	DP
	event-8	Generate a unique parts issue slip and gate pass for the engineer	DP
	event-9	Show “Inventory Issued successfully! Material slip generated” notification	OP
User-4, Goal-4	Description of Goal-4: Offer Tender		
Workflow:	event-1	events of login process to be executed	See login process

	event-2	If login is successful, load “Inventory Manager” dashboard	OP
	event-3	Select “Offer Tender” option	UIE
	event-4	Enter tender details: - Title (Supply of heavy fuel, Spare part of turbines or other equipments) - Technical specifications - Quantity - Estimated budget	UID
	event-5	Verify if provided data is valid	VL, VR
	event-6	If verification of event-5 is successful: - Show Tender on dashboard - Show “Tender Created!” notification Else: - Goto event-4	OP
User-4, Goal-5	Description of Goal-5: Vendor Management		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “Inventory Manager” dashboard	OP
	event-3	Select “Vendor Management” option	UIE
	event-4	Enter supply details: - Vendor name - Vendor address - Equipments offered	UID
	event-5	Verify if the provided data is valid	VL
	event-6	If verification of event-5 is successful: - Save supply information - Generate vendor ID - Show “Vendor information added” Else: - “Failure data!” warning - Goto event-4	DP, OP

User-4, Goal-6	Description of Goal-6: Inventory Report		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Inventory Report" option	UIE
	event-4	Select criteria for report generation: - Report type (Stock valuation, Fuel Consumption, Damaged items) - Date range (Weekly, Monthly, Yearly) - Power plant unit	UIE, UID
	event-5	Generate report	DP
User-4, Goal-7	Description of Goal-7: Search Equipment		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Search Equipment" option	UIE
	event-4	Fillup equipment details: - ID - Name - Category	UID
	event-5	Verify if equipment available	VR, VL
	event-6	If verification of event-5 is successful show details for each equipment: - Quantity available - Functional state - Last maintenance date	OP

		Else: - Show "No matching equipment found!" notification	
User-4, Goal-8	Description of Goal-8: Tracking Fuel Stock		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Tracking Fuel Stock" option	UIE
	event-4	Shows live quantity of fuel: - Fuel Type - Daily Consumption rate - Current Reserved quantity	OP
	event-5	Checks if current reserved quantity is lower than the threshold	DP
	event-6	If quantity is low: - Shows "Low Fuel Stock!" warning	OP

User-5 name: CEO			
User-5, Goal-1	Description of Goal-1: View plant performance summary		event type
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Plant Performance" menu.	UIE
	event-4	The system fetches today's total power generation (MW).	DP
	event-5	The system shows total MW output and plant efficiency (%).	OP
	event-6	Admin selects date range (e.g., last 7 days) to compare output.	UID
	event-7	The system fetches historical generation data and displays a trend chart.	DP, OP
User-5, Goal-2	Description of Goal-2: Check financial summary (profit/loss)		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Financial Summary" button.	UIE
	event-4	The system fetches revenue, fuel cost, and operational cost from records.	DP
	event-5	The system shows total revenue, total cost, and net profit.	OP

User-5, Goal-3	Description of Goal-3: Set or change tariff rates		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Tariff Settings" menu.	UIE
	event-4	Admin enters a new rate (e.g., 12 BDT/kWh for industrial).	UID
	event-5	The system checks that the rate is not zero or negative.	VL
	event-6	The system shows "Tariff updated successfully".	OP
User-5, Goal-4	Description of Goal-4: Approve fuel purchase request		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Pending Fuel Requests" tab.	UIE
	event-4	The system fetches a list of requests from the Plant Manager.	DP
	event-5	Admin clicks "Approve" on a request.	UIE
	event-6	System shows "Request approved. Order placed." message.	OP
User-5, Goal-5	Description of Goal-5: Add or remove an employee		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "User Management" menu.	UIE

	event-4	CEO enters employee name, role, and login ID.	UID
	event-5	System checks that the login ID is not already used.	VR
	event-6	The system shows "Employee added successfully".	OP
User-5, Goal-6	Description of Goal-6: View total registered customers		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Customer Summary" menu.	UIE
	event-4	The system fetches a total number of registered customers.	DP
	event-5	The system shows the count of industrial, commercial, and residential customers.	OP
User-5, Goal-7	Description of Goal-7: View customer complaint summary		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Complaint Summary" button.	UIE
	event-4	The system fetches total complaints (resolved/pending) from customer records.	DP
	event-5	The system shows the number of complaints today, this week, and still pending.	OP
User-5, Goal-8	Description of Goal-8: Generate weekly report		
<u>Workflow:</u>	event-1	Login process	CHECK

			LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	CEO clicks the "Generate Weekly Report" button.	UIE
	event-4	The system fetches generation, fuel, and financial data for the last 7 days.	DP
	event-5	The system calculates total output, total cost, and net profit.	DP
	event-6	The system displays reports on screen.	OP

User-6 name: Plant Manager

User-6, Goal-1	Description of Goal-1: Monitor current power generation		event type
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	Plant Manager clicks the "View Generation" tab.	UIE
	event-4	System fetches current MW output of all units.	DP
	event-5	The system shows MW output for Unit 1, Unit 2, Unit 3.	OP
User-6, Goal-2	Description of Goal-2: Check fuel level		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Customer Service Officer dashboard	OP
	event-3	Plant Manager clicks the "Fuel Status" menu.	UIE
	event-4	The system fetches current fuel stock (diesel/gas).	DP
	event-5	The system shows fuel level in liters and days remaining.	OP
User-6, Goal-3	Description of Goal-3: Schedule maintenance for a unit		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS

	event-2	If login is successful, load Customer Service Officer dashboard	OP
	event-3	Plant Manager clicks "Schedule Maintenance".	UIE
	event-4	Plant Manager selects units and enters date & time.	UID
	event-5	The system checks if that time is available.	VL
	event-6	System shows "Maintenance scheduled successfully".	OP
User-6, Goal-4	Description of Goal-4: Start or stop a generating unit		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Customer Service Officer dashboard	OP
	event-3	Plant Manager selects a unit (e.g., Unit 2).	UIE
	event-4	Plant Manager clicks the "Start" or "Stop" button.	UIE
	event-5	System checks if action is allowed (e.g., fuel available to start).	VL
	event-6	The system shows a "Unit started" or "Unit stopped" message.	OP
User-6, Goal-5	Description of Goal-5: View maintenance history		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Customer Service Officer dashboard	OP
	event-3	Plant Manager clicks "Maintenance History".	UIE
	event-4	The Plant Manager selects a unit number.	UID
	event-5	The system fetches past maintenance records.	DP

	event-6	The system shows a list of previous maintenance dates and work done.	OP
User-6, Goal-6	Description of Goal-6: Generate daily shift report		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Customer Service Officer dashboard	OP
	event-3	Plant Manager clicks "Generate Report".	UIE
	event-4	Plant Manager selects date and shift (Morning/Evening/Night).	UID
	event-5	The system fetches generation and fuel data for that shift.	DP
	event-6	The system shows total power generated and fuel used.	OP
User-6, Goal-7	Description of Goal-7: Raise fuel request to Admin		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Customer Service Officer dashboard	OP
	event-3	Plant Manager clicks the "Request Fuel" button.	UIE
	event-4	Plant Manager enters fuel type (diesel/gas) and quantity.	UID
	event-5	The system checks that quantity is a positive number.	VL
	event-6	The system shows "Fuel request sent to Admin for approval".	OP

User-6, Goal-8	Description of Goal-8: Log an unplanned breakdown		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Customer Service Officer dashboard	OP
	event-3	Plant Manager clicks "Log Breakdown".	UIE
	event-4	Plant Manager selects units, enters time, and writes reasons.	UID
	event-5	The system checks that all fields are filled.	VL
	event-6	System shows "Breakdown reported to Admin".	OP

User-7 name: Engineer			
User-7, Goal-1	Description of Goal-1: Monitor Generator Status		event type
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Engineer Dashboard, where the engineer can access Generator Monitoring, Maintenance Management, Fault Reporting, Alarm Management, and Performance Reports	OP
	event-3	Select Generator Monitoring option.	UIE
	event-4	Select the Generator ID or Plant Unit to monitor. The system displays: • Generator ID • Plant Unit • Running Status • Current Power Output (MW) • Voltage • Frequency • Fuel Level • Engine Temperature • Running Hours • Last Maintenance Date	UID
	event-5	Verify whether the generator exists and all operating values are received correctly from the system database. Also verify whether the generator is operating within the predefined safety limits.	VL,VR
	event-6	If verification of event-5 is successful: Display the generator's operational status. If any abnormal value is detected, display "Generator Warning! Immediate Attention Required." Else: Show "Unable to retrieve generator information."	DP,OP

		warning. Goto event-4.	
User-7, Goal-2	Description of Goal-2: Schedule Preventive Maintenance		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Maintenance Schedule option	UIE
	event-4	Fill up maintenance details: • Equipment ID • Generator Unit • Maintenance Type (Routine/Emergency) • Scheduled Date • Assigned Technician • Expected Duration • Remarks	UID
	event-5	Verify whether the equipment exists, all required fields are completed, and the selected maintenance date does not conflict with another scheduled maintenance	VL,VR
	event-6	If verification of event-5 is successful: Generate Maintenance Schedule ID. Save maintenance information. Display "Maintenance Scheduled Successfully." Else: Show "Invalid Maintenance Information!" warning. Goto event-4	DP,OP
User-7, Goal-3	Description of Goal-3: Report equipment faults		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Report Equipment Fault option	UIE
	event-4	Fill up fault details: • Equipment ID • Generator Unit	UID

		• Fault Category • Fault Severity (Low/Medium/Critical) • Date & Time • Fault Description	
	event-5	Verify whether the Equipment ID exists and all mandatory information has been provided correctly.	VL,VR
	event-6	If verification of event-5 is successful: Generate a unique Fault Report ID. Save fault information. Notify the Maintenance Department. Show "Fault Report Submitted Successfully." Else: Show "Invalid Equipment Information!" warning. Goto event-4.	DP,OP
User-7, Goal-4	Description of Goal-4: Update Maintenance Status		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Update Maintenance Status option.	UIE
	event-4	Enter maintenance information: • Maintenance ID • Equipment ID • Completion Date • Parts Replaced • Engineer Remarks • Current Equipment Condition	UID
	event-5	Verify whether the Maintenance ID exists and all required maintenance information is valid.	VL,VR
	event-6	If verification of event-5 is successful: Update maintenance record. Change equipment status to Operational or Under Observation. Show "Maintenance Record Updated Successfully." Else: Show "Maintenance Record Not Found!" warning. Goto event-4.	DP,OP

User-7, Goal-5	Description of Goal-5: Analyze equipment efficiency		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Performance Report option.	UIE
	event-4	Select report criteria: • Plant Unit • Generator ID (Optional) • Report Type (Power Generation, Fuel Consumption, Equipment Efficiency, Downtime) • Date Range (Daily, Weekly, Monthly, Yearly)	UID
	event-5	Verify whether the selected report criteria are valid and data is available for the requested period.	VL,VR
	event-6	If verification of event-5 is successful: Generate performance report. Display generation statistics, fuel usage, efficiency, downtime, and maintenance summary. Else: Show "No Report Available." notification. Goto event-4.	DP,OP
User-7, Goal-6	Description of Goal-6: Monitor fuel consumption		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Search Equipment option.	UIE
	event-4	Enter equipment information: • Equipment ID • Equipment Name • Generator Unit	UID
	event-5	Verify whether the equipment exists in the system database.	VL,VR
	event-6	If verification of event-5 is successful, display: • Equipment Status	DP,OP

		• Installation Date • Last Maintenance Date • Next Maintenance Date • Current Condition Else: Show "No Matching Equipment Found!" notification.	
User-7, Goal-7	Description of Goal-7: Manage spare parts requirements		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Alarm Management option.	UIE
	event-4	View active alarm details including: • Alarm ID • Equipment ID • Generator Unit • Alarm Type • Severity Level • Time of Occurrence	OP
	event-5	Verify whether the alarm is active and has not already been resolved by another engineer.	VL,VR
	event-6	If verification of event-5 is successful: Update alarm status (Acknowledged/Resolved). Enter corrective action taken. Save response history. Show "Alarm Updated Successfully." Else: Show "Alarm Already Resolved." notification.	DP,OP
User-7, Goal-8	Description of Goal-8: Generate engineering performance reports		
Workflow:	event-1	Events of login process to be executed. See Login Process	See login process
	event-2	If login is successful, load Engineer Dashboard	OP
	event-3	Select Daily Operation Log option	UIE

	event-4	Fill up daily operation details: • Shift Name • Generator Unit • Generated Power (MW) • Fuel Consumed • Operating Hours • Equipment Condition • Engineer Remarks	UID
	event-5	Verify whether all mandatory fields are completed and the entered operational values are valid	VL,VR
	event-6	If verification of event-5 is successful: Generate Daily Log ID. Save operation log. Display "Daily Operation Log Recorded Successfully." Else: Show "Invalid Operation Data!" warning. Goto event-4	DP,OP

User-8 name: Grid Operator			
User-8, Goal-1	Description of Goal-1: Monitor Grid Status		event type
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard, displaying the overall grid status, active power generation, connected generator units, transmission status, active alarms, and grid frequency.	OP
	event-3	Select Grid Monitoring option.	UIE
	event-4	Select the power plant unit or transmission line to monitor. The system displays: • Grid Frequency	UID

		• Grid Voltage • Active Power (MW) • Reactive Power (MVAR) • Load Demand • Transmission Line Status • Connected Generator Units	
	event-5	Verify whether the selected grid data is available and whether all operating parameters are within the predefined operating limits.	VL,VR
	event-6	If verification of event-5 is successful: • Display the real-time grid status. • Highlight abnormal values if any parameter exceeds the safety limit. • Show "Grid Operating Normally." if all values remain within acceptable limits. Else: • Show "Unable to retrieve grid information!" warning. • Goto event-4.	DP,OP
User-8, Goal-2	Description of Goal-2: Monitor Load Demand		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Load Demand Monitoring option.	UIE
	event-4	Select monitoring criteria: • Plant Unit • Region • Date & Time The system displays: • Current Load Demand • Available Generation Capacity	UID

		• Peak Load • Reserve Capacity	
	event-5	Verify whether demand information is available for the selected criteria.	VL,VR
	event-6	If verification of event-5 is successful: • Display current load demand and reserve capacity. • If demand exceeds generation capacity, show "High Load Demand! Additional Generation Required." Else: • Show "Load information unavailable!" • Goto event-4.	DP,OP
User-8, Goal-3	Description of Goal-3: Synchronize Generator with National Grid		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Generator Synchronization option.	UIE
	event-4	Enter synchronization details: • Generator ID • Plant Unit • Grid Frequency • Generator Frequency • Synchronization Time	UID
	event-5	Verify whether the generator frequency, voltage, and phase sequence match the grid requirements for safe synchronization.	VL,VR
	event-6	If verification of event-5 is successful:	DP,OP

		• Synchronize the generator with the national grid. • Update generator status to Connected. • Display "Generator Successfully Connected to Grid." Else: • Show "Synchronization Failed! Frequency or Voltage Mismatch." • Goto event-4.	
User-8, Goal-4	Description of Goal-4: Respond to Grid Fault		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Grid Fault Management option.	UID
	event-4	View fault details including: • Fault ID • Transmission Line • Fault Type • Time of Occurrence • Affected Plant Unit • Severity Level	VL
	event-5	Verify whether the fault is active and has not already been resolved.	VR
	event-6	If verification of event-5 is successful: • Record corrective action taken. • Update fault status. • Notify the Control Center. • Show "Fault Response Recorded Successfully." Else:	DP

		• Show "Fault Already Resolved."	
User-8, Goal-5	Description of Goal-5: Schedule Load Distribution		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Load Distribution option.	UIE
	event-4	Enter distribution details: • Plant Unit • Region • Required Power (MW) • Distribution Time • Priority Level	UID
	event-5	Verify whether sufficient generation capacity is available for the requested load distribution.	VL,VR
	event-6	If verification of event-5 is successful: • Save load distribution schedule. • Update generation allocation. • Display "Load Distribution Scheduled Successfully." Else: • Show "Insufficient Generation Capacity!" • Goto event-4.	DP,OP
User-8, Goal-6	Description of Goal-6: View Grid Performance Report		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP

	event-3	Select Grid Performance Report option.	UIE
	event-4	Select report criteria: • Report Type • Plant Unit • Region • Date Range	UID
	event-5	Verify whether report data exists for the selected period.	VL,VR
	event-6	If verification of event-5 is successful: Display: • Power Supplied • Peak Load • Average Frequency • Grid Stability • Outage Summary Else: • Show "No Report Available."	DP,OP
User-8, Goal-7	Description of Goal-7: Search Transmission Line Information		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Search Transmission Line option.	UIE
	event-4	Enter: • Transmission Line ID • Region • Substation Name	UID
	event-5	Verify whether the transmission line exists in the database.	VL,VR

	event-6	If verification of event-5 is successful display: • Line Status • Voltage Level • Connected Substations • Current Load • Last Inspection Date Else: • Show "No Matching Transmission Line Found!"	DP,OP
User-8, Goal-8	Description of Goal-8: Record Grid Operation Log		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Grid Operation Log option.	UIE
	event-4	Fill up operation details: • Shift • Plant Unit • Grid Frequency • Total Power Supplied • Load Demand • Major Events • Operator Remarks	UID
	event-5	Verify whether all mandatory fields have been completed and the operational values are valid	VL,VR
	event-6	If verification of event-5 is successful: • Generate Grid Log ID. • Save operation log. • Display "Grid Operation Log Recorded Successfully."	DP,OP

		Else: • Show "Invalid Operation Data!" • Goto event-4.	
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